# The empirical recurrence for OEIS A183730, proved by transfer matrix

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**Abstract**

OEIS A183730 counts arrays of width 2 over $\{0, 1, \ldots, 7\}$ subject to a condition read round the perimeter of every $2 \times 2$ subblock: every subblock turns strictly one way with a single decrease, and some two adjacent subblocks turn the same way. The entry carries an empirical recurrence of order 9 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: the condition is settled inside a bounded window of consecutive rows, so the count is a walk count on a finite digraph with $S = 225$ vertices, and whether a linear recurrence holds for such a count is a finite computation.

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## 1 The conjecture

OEIS A183730 is "1/16 the number of (n+1) X 2 0..7 arrays with every 2 X 2 subblock strictly increasing clockwise or counterclockwise with one decrease, and at least two adjacent blocks having the same increasing direction.". It has offset 1 and begins

$$0, 21, 563, 9403, 135578, 1814815, 23204876, 287769879, \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n)=18*a(n-1)-46*a(n-2)-342*a(n-3)-33*a(n-4)+435*a(n-5)-157*a(n-6)-37*a(n-7)+21*a(n-8)-2*a(n-9).

As of the "Last modified" line on the live entry (Oct 06 2015 22:06:56, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

## 2 What is being counted

The arrays have 2 columns and a growing number of rows, with entries in $\{0, 1, \ldots, 7\}$. A $2 \times 2$ subblock is a set of four cells

$$B = \begin{pmatrix} p & q \\ r & s \end{pmatrix}, \qquad p = x(i, j),\ q = x(i, j+1),\ r = x(i+1, j),\ s = x(i+1, j+1).$$

Its perimeter is a cycle of four cells, walked clockwise as $p \to q \to s \to r \to p$ and counterclockwise as $p \to r \to s \to q \to p$. Write

$$\mathrm{cw}(B) = [p < q] + [q < s] + [s < r] + [r < p], \qquad \mathrm{ccw}(B) = [p < r] + [r < s] + [s < q] + [q < p]$$

for the number of steps at which the value rises, going each way round.

**Lemma 1.** $\mathrm{cw}(B) + \mathrm{ccw}(B) = 4 - \#\{$*edges of the cycle with equal ends*$\}$.

*Proof.* Each of the four edges of the square is walked once by each cycle, and in opposite directions. An edge with distinct ends therefore contributes exactly one rise, to one of the two counts; an edge with equal ends contributes to neither. □

The condition the entry imposes is this: every subblock $B$ has its four corners pairwise unequal along the cycle ($p \neq q$, $q \neq s$, $s \neq r$, $r \neq p$) and

$$\mathrm{cw}(B) \in \{1, 3\}.$$

Going round clockwise there are then four strict steps, of which either three rise and one falls ($\mathrm{cw}(B) = 3$: strictly increasing clockwise with one decrease) or one rises and three fall ($\mathrm{cw}(B) = 1$, which is the same statement read counterclockwise). Call the subblock *clockwise* in the first case and *counterclockwise* in the second; this is its increasing direction.

The entry asks in addition that at least two adjacent subblocks —adjacent meaning one step right or one step down in the grid of subblocks—have the same increasing direction. That is a condition on the array as a whole, not on any one subblock. The entry publishes 1/16 of that count; the division is by the constant 16 and changes nothing about the recurrence.

## 3 A bounded window

Two consecutive rows decide every subblock lying between them, and their increasing directions. The extra requirement is an EXISTENCE statement about the whole array, and an existence statement needs only one bit: the state carries the row just written, the increasing direction of each subblock of the pair just closed—which is what a subblock in the next pair must be compared against—and a single flag, set the first time two adjacent subblocks are seen to agree and never cleared.

One step of the walk writes a whole row. The row is filled left to right, and the subblock closed by positions $j-1, j$ is tested the moment position $j$ is written, so a partial row that cannot be completed is abandoned rather than enumerated. The walk starts at the state standing for the empty array, so that a walk of $n$ steps is an array of $n$ rows.

The end vector is the indicator of the states in which the required pair of adjacent subblocks has already been witnessed; every other state is rejected. Thus

$$a(n) \;=\; \iota^\top M^{\,j} \tau, \qquad S = 225,$$

for the transfer matrix $M$ on those $S$ states. The walk index $j$ and the entry's own $n$ differ by a constant, read off by matching the published terms rather than assumed. In particular $a$ is $C$-finite, and every question about linear recurrences it satisfies is decidable.

## 4 The proof

**Lemma 2.** *Let $q(t) = t^9 - \sum_i c_i t^{9-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^{\,j} q(M)\tau$ for the corresponding walk index $j$, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large $j$, and it is enough to examine $j < S$.*

*Proof.* Factor $M^j$ out of $M^{j+9} - \sum_i c_i M^{j+9-i}$. For the bound, $M^S$ is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of $M$; $S$ consecutive zeros force every later one, and the last nonzero $u_j$ pins the threshold exactly. □

**Theorem 3.**

$$a(n) = 18\,a(n-1)-46\,a(n-2)-342\,a(n-3)-33\,a(n-4)+435\,a(n-5)-157\,a(n-6)-37\,a(n-7)+21\,a(n-8)-2\,a($$

*for every $n > 9$.*

*Proof.* The vector $w = q(M)\tau$ was formed by 9 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 9 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger $j$ at once. □

## 5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: every array of the stated width and a few rows was written out cell by cell over $\{0, 1, \ldots, 7\}$, each of its $2 \times 2$ subblocks was read round its perimeter, and the condition of Section 2 was applied as stated. No window, no state and no recurrence is involved in that computation, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

## References


[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A183730.

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[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.